

Data Cleaning Change Log

Project 1 — Data Cleaning & Preparation | DecodeLabs Internship, Batch 2026

Dataset: Dataset_for_Data_Analytics.xlsx → Dataset_for_Data_Analytics_CLEANED.xlsx

Rows: 1,200 | Columns: 14 | Unique key: OrderID

Change Record

Change ID	Description	Impact	Status
CR001	Identified 309 missing values in CouponCode (25.75% of rows). Confirmed these represent orders placed without a coupon, not a logging error, by cross-checking against TotalPrice. Imputed missing values with the explicit label 'No Coupon'.	Preserved all 1,200 records (no rows dropped). CouponCode is now a complete, analysis-ready categorical field.	Resolved
CR002	Audited full dataset for duplicate records using both a full-row comparison and a check on the OrderID unique key.	0 duplicate rows found; 0 duplicate OrderIDs found. OrderID confirmed valid as a unique identifier. No rows removed.	Verified – No Action Needed
CR003	Standardized the Date column to ISO 8601 format (YYYY-MM-DD) to remove any ambiguity in how dates are stored or reopened in Excel.	All 1,200 Date values now conform to a single, unambiguous format.	Resolved
CR004	Detected floating-point precision errors in TotalPrice introduced by the Quantity × UnitPrice calculation (e.g. a displayed value of 769.38 was internally stored as 769.37999999999988). Rounded UnitPrice and TotalPrice to exactly 2 decimal places.	Eliminated hidden floating-point noise in 29 affected rows. All monetary values now carry consistent 2-decimal precision.	Resolved
CR005	Validated business logic: confirmed Quantity × UnitPrice = TotalPrice on every row, and checked for negative values across Quantity, UnitPrice, and TotalPrice.	0 calculation mismatches; 0 negative values found. TotalPrice confirmed reliable for downstream revenue analysis.	Verified – No Action Needed
CR006	Investigated disagreement between Quantity and ItemsInCart (1,000 of 1,200 rows differed). Found ItemsInCart is never less than Quantity, consistent with ItemsInCart representing total cart size rather than a duplicate measurement.	No correction applied. Documented as an expected difference between two distinct, valid fields.	Verified – No Action Needed
CR007	Checked all categorical columns (Product, PaymentMethod, OrderStatus, ReferralSource) for inconsistent casing, typos, or stray whitespace.	All categories confirmed clean and consistent. No standardization required.	Verified – No Action Needed

Verification Gate (Project 2 Threshold)

Requirement	Result
0% error rate on unique identifiers (OrderID)	Passed — 0 duplicates found
0% error rate on date formats	Passed — all dates standardized to YYYY-MM-DD